

Eventra

A Web-Based University Event and Club Management Platform

Software Engineering and Information System Design Lab

Course Code: CSE-3104

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Team Members and Roles

Name	Student ID	Role & Responsibilities
MD. Tanvir Ahmed Rana	230102016	Team Leader – Project coordination, task distribution, requirement analysis, documentation, SDLC planning, Git collaboration, HTML, CSS, PHP, and overall project supervision.
MD. Readul Haque Sohan	230102015	Backend Developer – Laravel backend development, authentication, role-based access, event management logic, registration workflow, and API/module integration.
Rabea Amrin Linat	230102047	Database Administrator – Database schema design, ERD preparation, normalization, MySQL table relationship design, query optimization, and data security planning.
Niloy Deb	230102030	UI/UX Designer and Frontend Developer – Interface design, responsive layout, event card design, dashboard UI, HTML, CSS, JavaScript, and Bootstrap/Tailwind implementation.
Syeda Fazilatunnassa	230102011	SQA Engineer – Test planning, functional testing, system testing, bug tracking, validation of registration, QR attendance, certificate, and dashboard modules.
Sarmin Sultana Ritu	230102006	Business Analyst and Documentation Coordinator – Problem analysis, market and user research, feature planning, report writing, budget estimation, timeline preparation, and final documentation support.

Project Idea

Title

Eventra – A Web-Based University Event and Club Management Platform

Overview

Eventra is an innovative web-based platform designed to digitize and centralize the complete lifecycle of university event and club management. It is specially designed for university students, clubs, departments, event organizers, and administrators.

Through Eventra, students can discover upcoming university events in a clean card-based interface, follow clubs, register for events, receive QR-based event passes, view event details, attend events through QR check-in, submit feedback, and download verified digital certificates. The system focuses on simple event discovery, category-based browsing, community engagement, and easy participation.

Club managers and organizers can create events with full details including event banner, category, description, date, time, venue, seat limit, registration deadline, and participation rules. They can manage participants, approve or reject registrations, track attendance, generate certificates, analyze feedback, and produce event impact reports. Administrators maintain complete oversight of users, clubs, event categories, approvals, analytics, and support messages from a centralized dashboard.

Eventra transforms the traditional university event workflow – previously dependent on paper forms, spreadsheets, social media posts, and manual attendance sheets – into a modern, organized, data-driven, and university-focused digital event management experience.

Problem Statement

In universities across Bangladesh, event and club management processes remain largely manual and fragmented. Event organizers commonly use physical registration forms, Google Forms, spreadsheets, Facebook posts, Messenger groups, email announcements, and manual attendance sheets to manage events. These disconnected tools create several problems such as duplicate registrations, disorganized participant data, lack of proper event discovery, attendance errors, delayed certificate distribution, weak feedback collection, and difficulty in generating event reports.

Students often miss important events because there is no centralized event discovery platform. They may not know which clubs are active, which events match their interests, or whether their registration has been approved. Organizers face difficulties managing large participant lists, verifying attendance, communicating updates, and distributing certificates on time. University administrators also lack a consolidated view of club activities, event performance, participation statistics, and student engagement. Generic platforms like social media event pages or public event listing websites are not designed for university-specific workflows. They do not properly support campus-based club approval, role-based dashboards, student verification, QR attendance, auto certificate generation, and event impact reporting. Therefore, there is a need for

a specialized university-focused platform that supports event discovery and academic event management features.

Eventra addresses this gap by providing a centralized web-based platform where the full lifecycle of a university event can be managed – from event creation and approval to registration, attendance, feedback, certificate generation, and final reporting.

Motivation

The motivation behind Eventra comes from the need to make university events more discoverable, organized, and engaging. As students, we have observed that many useful seminars, workshops, competitions, club programs, and career events fail to reach the right participants because event information is scattered across different channels. Even when students register, attendance and certificate distribution are often managed manually, causing confusion and delays.

Eventra aims to provide a campus-focused event platform where students can easily discover relevant events, clubs can manage activities professionally, and administrators can monitor campus engagement effectively.

Eventra is not only a software project; it is a digital solution for improving campus culture, student participation, club management, and event transparency.

Feature and Design Direction

Eventra will follow a modern, clean, and user-friendly design approach while remaining customized for university event and club management. The system will make events easy to discover, explore, and join.

Design Direction

- Card-based event listing with event banner, title, date, time, venue, category, and organizer name.
- Clean homepage with featured events, upcoming events, popular clubs, and event categories.
- Search and filter options based on event category, date, club, keyword, and participation type.
- Event detail page with clear information, registration button, seat availability, organizer details, and event schedule.
- Club profile pages showing club description, logo, upcoming events, past events, and member activity.
- Simple and responsive design suitable for desktop, tablet, and mobile screens.
- Dashboard-based interface for students, organizers, and admins.

Feature Direction

- Event discovery based on interest and category.
- Club or community-based event organization.

- Student ability to follow clubs and view their upcoming events.
- Recommended events based on student interests, department, and previous participation.
- Registration system where students can request to join an event.
- Event reminder and notification system.
- Participant list and attendance tracking for organizers.
- Post-event feedback and rating system to measure event quality.
- Event archive showing previous events and participation records.

University-Specific Customization

- Admin approval before events become public.
- QR-based event pass and attendance verification.
- Auto certificate generation after successful attendance.
- Certificate verification using unique certificate codes.
- Seat limit and waitlist management for high-demand events.
- Event impact report for clubs and university administration.
- Role-based access for Admin, Organizer, Student, and Guest users.

SWOT Analysis

Strengths

- University-focused event discovery model adapted specifically for campus event management.
- The team directly understands the real challenges in university event management and participant coordination.
- Balanced skillset across UI/UX design, backend development, database administration, and quality assurance.
- Strong motivation rooted in solving a real, observable problem in the campus environment.
- Combination of event discovery, club management, QR attendance, certificate generation, and analytics in one platform.

Weaknesses

- Limited development time due to academic coursework and examination schedules.
- Heavy course load may affect the speed of feature implementation and testing cycles.

- Dependence on key team members for critical modules such as QR generation, certificate automation, and recommendation features.
- Advanced recommendation features may need to be implemented in a simplified form during the initial version.

Opportunities

- High educational and administrative impact across universities lacking digital event management tools.
- Potential to become a central event discovery and club activity platform for university students.
- Fully web-based system requires no hardware investment, enabling quick adoption.
- Potential for collaboration with university administration, student clubs, and event committees.
- Future expansion to mobile apps, AI-based event recommendation, online ticketing, and multi-university support.

Threats

- Competition from generic tools such as Google Forms, social media events, and public event platforms.
- User adoption challenges, particularly among students and organizers used to manual or social media-based processes.
- Security and trust concerns around QR-based attendance and certificate verification.
- Possible server and connectivity issues affecting real-time QR scanning and data operations.

Expected Features / Requirements

Student / Participant Profile

- Register an account and log in securely.
- Create and update student profile with department, batch, student ID, and interests.
- Browse upcoming university events in a card-based layout.
- Search and filter events by category, date, club, keyword, and interest.
- Follow clubs and view their upcoming and past events.
- Register for events and track registration status.
- Receive a QR-based digital event pass upon organizer approval.
- Attend events using the QR pass for digital check-in.

- Submit post-event feedback and star ratings.
- Download auto-generated participation certificates.
- Verify certificates using a unique verification code.
- Receive system notifications for approvals, reminders, event updates, and certificate availability.

Organizer / Club Manager Profile

- Log in to the organizer dashboard.
- Create, edit, and delete club events with full details: banner, category, description, date, time, venue, seat limit, registration deadline, and rules.
- Manage club profile with logo, description, category, and contact information.
- View and manage registered participants – approve, reject, or waitlist registrations.
- Scan participant QR passes or manually mark attendance.
- Generate and issue certificates for eligible attendees.
- Send event updates and reminders to approved participants.
- View post-event feedback and ratings submitted by participants.
- Generate and download event impact reports.
- View club performance based on total events, participants, attendance rate, feedback rating, and certificate count.

Admin Profile

- Manage all user accounts across roles.
- Manage clubs – create, approve, update, or deactivate clubs.
- Approve or reject events submitted by organizers before public listing.
- Manage event categories for consistent classification.
- Monitor platform-wide statistics: total events, active clubs, participants, certificates issued, and feedback received.
- View club performance analytics.
- Generate system-wide reports and analytics.
- Manage and respond to contact and support messages.
- Monitor system security, user status, and platform activity.

Guest / Visitor Access

- View the homepage, public event listings, and event details.
- Browse registered clubs and their profiles.
- Search public events by category, date, and keyword.

- Access the contact/support page.
- Register for an account or log in to access full features.

Special Features

- Modern event discovery interface with card-based event listings.
- Interest-based event recommendation for students.
- Club following system so students can track activities of their favorite clubs.
- QR code-based event pass generation and attendance verification.
- Event registration with approval, rejection, and waitlist options.
- Automatic certificate generation for eligible participants.
- Certificate verification system using unique verification codes.
- Role-based dashboards for Admin, Organizer, and Student.
- Seat and waitlist management with registration deadline enforcement.
- Event reminder and notification system.
- Event impact reporting with participation statistics, attendance percentage, feedback rating, and certificate count.
- Club performance analytics for university administrators.
- Digital event archive for completed events.
- Email OTP verification and password recovery.

Development Approach

Model: Agile

1. Planning & Requirement Analysis
2. Design (UI/UX & System Architecture)
3. Implementation / Development
4. Testing & Quality Assurance
5. Deployment
6. Maintenance & Feedback

The Agile model is selected to support iterative development, frequent testing, and continuous feedback integration throughout the project lifecycle. Each sprint will focus on delivering a functional part of the system, such as authentication, event discovery, club management, registration, QR attendance, certificate generation, or reporting.

The development process will place special emphasis on user experience, event browsing flow, responsive event cards, search and filter functionality, and smooth interaction between students and clubs. Feedback from users and supervisors will be collected after each major module and used to improve the system.

Tools and Technologies

Category	Recommended Tools	Purpose / Justification
Frontend	HTML5, CSS3, Bootstrap / Tailwind CSS, JavaScript	Responsive user interface, event cards, club pages, dashboards, filters, and dynamic interactions.
Backend	PHP (Laravel Framework)	Clean MVC structure with authentication, routing, authorization, notifications, and event management logic.
Database	MySQL	Suitable for relational data such as users, clubs, events, registrations, attendance, feedback, and certificates.
Design	Figma / Adobe XD	UI/UX prototyping for homepage, event cards, club profiles, event details, and dashboards.
QR Code	SimpleSoftware/php-qrcode	QR pass generation and scanning for event attendance verification.
Certificate	DomPDF / TCPDF	Automated PDF certificate generation from Laravel templates.
Testing	PHPUnit, Selenium	Unit testing for backend logic and UI testing for registration, search, QR, and certificate flows.
Version Control	Git + GitHub	Team collaboration, version tracking, and code review.
Deployment	cPanel / AWS EC2	Standard PHP hosting setup for deployment.
Management	Trello / Notion / Google Meet	Task tracking, sprint planning, meeting notes, and team coordination.
Notifications	PHPMailer / Laravel Notifications	Email OTP, registration alerts, event reminders, and certificate notifications.
Visualization	Chart.js / Google Charts	Admin analytics dashboard, club performance charts, and event impact reports.
Security	SSL + Laravel Auth + password hashing	Safe authentication, role-based access, user data protection, and certificate verification security.

Budget Estimation

Development Costs (One-Time)

Category	Description	Min (BDT)	Max (BDT)	Notes
Frontend	Responsive UI, event cards, dashboards, event pages, club profiles, QR pass views	25,000	45,000	One-Time
Backend	Laravel APIs, event discovery logic, QR logic, certificate engine, auth system	30,000	60,000	One-Time
Database	Schema design, indexing, migration scripts, relationship design for clubs and events	10,000	20,000	One-Time
Optional	Admin analytics, reporting module, notification system, recommendation logic	15,000	40,000	One-Time
Subtotal		80,000	1,65,000	

Additional One-Time Costs

Category	Description	Min (BDT)	Max (BDT)	Notes
Domain Name	.com/.edu, 1–2 years	850	2,000	One-Time
SSL Certificate	Free / Premium	0	3,000	Often free
Content Writing	About, FAQ, Help, event rules, and public pages	3,000	8,000	One-Time
SEO & Marketing	Meta tags, analytics, launch promotion, event discovery optimization	5,000	15,000	One-Time
Logo & Graphics	Logo, event banners, favicon, certificate template, club placeholders	2,000	10,000	One-Time

Hosting Options & Annual Costs

BDIX Hosting (Budget-Friendly)

Provider	Package	Storage	Bandwidth	Cost/Year (BDT)
HostMight	Pro	20GB SSD	Unlimited	3,500

Table 1: BDIX Hosting Options and Annual Costs

Timeline

Duration	Tasks
Days 1–5	Project Requirement Analysis and Planning; studying university event management needs and event discovery flow.
Days 6–15	UI/UX Design – homepage, event cards, club profiles, event details, dashboards, Figma mockups, and design system setup.
Days 16–25	Database Schema Design, ERD, Laravel Project Initialization, and user role structure setup.
Days 26–38	Backend Development: Authentication, Events, Clubs, Registration, Follow Club, QR Logic, and Approval Workflow.
Days 39–50	Frontend Integration: Event browsing, Student Dashboard, Organizer Dashboard, Admin Dashboard, and Laravel Blade pages.
Days 51–58	QR Attendance System, Certificate Generation, Feedback, Notification, Recommendation, and Reports Module.
Days 59–65	Testing, Bug Fixing, Security Review, Performance Optimization, and UI Responsiveness Testing.
Days 66–70	Deployment, Final Documentation, Project Report, Presentation Preparation, and Final Submission.

Conclusion

Eventra is designed to address one of the most common operational challenges faced by university campuses – the lack of a centralized, efficient, and engaging event and club management system. Eventra brings a modern event browsing and participation experience into the university environment.

Unlike a simple event listing website, Eventra manages the complete event lifecycle: event creation, admin approval, student registration, QR attendance, feedback collection, certificate generation, certificate verification, and event impact reporting. For students, it provides a simple way to discover events, follow clubs, register according to interests, attend with a QR pass, and receive verified certificates. For organizers and club managers, it offers a professional dashboard to manage events, participants, attendance, feedback, and reports. For administrators, it provides clear visibility into campus activities, club performance, and student engagement.

Future enhancements may include native mobile applications, AI-based event recommendations, online paid event registration, live streaming support for hybrid events, and multi-university deployment. The platform may also be expanded with advanced analytics to measure long-term student participation and club effectiveness.

In essence, Eventra represents a university-focused vision: making campus events easier to discover, easier to manage, and more meaningful for every student, organizer, and administrator.